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E-400 SUBSTATION AND APPURTENANCES

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to site, installation, commissioning and testing 500 kV conventional substation extension and appurtenances in accordance with Part 4 Subsection 1.2 Summary of Work and as shown on the drawings.

The work shall also include integration of this substation scope to the existing substation in all aspects such as in power, metering, protection, control, communication, SCADA including RTU, LFC, grounding, physical appearance, etc. in order to constitute proper 500 kV generating substation. The design shall also in line with the PLN Grid Code requirements.

The scope of work shall comprise, but not limited to, the following :

- (1) Three (3) complete **one and half breaker** generator line diameters No. 6, 11 and 12, including the complete integration to the existing substation system.
- (2) Complete and proper connection and function the output of the new steam turbines to diameters No. 6, 11 and 12 of the existing 500 kV Substation including all necessary hardware and complete integration to the existing system.
- (3) The Contractor shall investigate the existing system and provide all the necessary panels for the diameters No. 5, 6, 11 and 12 as it is control, monitoring, metering, protection, substation interface and DCS panels and also SCADA panels for Block 2,3 and 4, including all the necessary interface.
- (4) The Contractor shall investigate the existing AC/DC system and provide the necessary requirement power supply of the AC, DC and UPS for the equipment including all necessary civil works.

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2. CODE AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards, and the following specific standards:

2.1 Indonesian National Standards

PUIL : Peraturan Umum Instalasi Listrik-Indonesia (General Regulation of Electric Installation-Indonesia)

SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 International Electrotechnical Commission

IEC 60044 : Instrument transformers

IEC 60137 : Insulated bushings for alternating voltages above 1000 V

IEC 60186 : Voltage transformers

IEC 60376 : Specification and acceptance of new sulphur hexafluoride

IEC 60694 : Common specifications for high-voltage switchgear and controlgear standards

IEC 62155 : Hollow pressurized and unpressurized ceramic and glass insulators for use in electrical equipment with rated voltages greater than 1000 V

IEC 62271 : High-voltage switchgear and controlgear

2.3 National Electrical Manufacturers Association

NEMA 107 : Methods of Measurements of Radio Influence Voltage (RIV) of High Voltage Apparatus

3. OPERATING AND DESIGN CONDITIONS

The substation and appurtenances shall be designed for outdoor installation in a tropical climate along the seacoast.

Unless specified otherwise, the equipment shall be designed for operation on an effectively grounded system.

4. DETAILED REQUIREMENTS

4.1 General

The Contractor shall arrange the substation to allow for ease of operation, maintenance, and proper integration with the existing substation system without creating any disturbance and defect and abnormalities to the existing plant and future expansion. Sufficient space and access areas shall be provided to permit ready removal and reinstallation of each internal and external equipment and component.

Any failed component or gasket sealing system shall be capable of being removed and replaced without requiring excessive addition or removal of bracing, or de-energizing of any piece of equipment not within the failed

electrical circuit or compartment.

4.2 Ratings

The following are the general ratings that all components or sections of the substation shall meet the required criteria :

(1)	Nominal system voltage	500 kV	
(2)	Highest system voltage	550 kV	
(3)	System frequency	50 Hz	
(4)	Lightning impulse withstand voltage	1,550 kV peak	
(5)	Switching impulse withstand voltage	1,175 kV peak	
(6)	Power frequency withstand voltage (*wet)	680 kV	
(7)	Short time withstand current	50 kA	ADD1 -IIE -002
(8)	Normal current	3,150 A	
(9)	Insulator creepage distance.....	16,275 mm	
(10)	Auxiliary and control circuit ratings.....	As per Technical Specification E-450	

5. PERFORMANCE TESTING AND GUARANTEES

The Contractor shall perform type tests, routine tests and site tests as specified hereinafter. Type test report of the identical equipment can be submitted in lieu of the test.

The lightning impulse test voltage for the type and routine tests shall be 1800 kV for 500 kV.

The partial discharge test shall be performed in accordance with IEC 62271. The guaranteed partial discharge intensity shall be included with the Bid information.

The Radio Interference Voltage (RIV) measurement shall be made in accordance with NEMA 107. In making the RIV tests, radio noise readings shall be taken at various points so as to determine the break point. The applied phase-to-ground voltage for the RIV test shall be 335 kV for 500 kV substation. The maximum RIV level at 1 MHz shall be 2,500 μ volts.

High-voltage tests shall be performed at the site to check dielectric strength of the completely assembled equipment. These tests shall not replace the specified type tests or the routine tests. These tests shall be separate from the progressive voltage increase performed for the electrical conditioning of the equipment before commissioning. The Contractor shall submit a high-voltage test program which shall comply with the following minimum requirements:

- (1) The high-voltage tests shall be performed at the end of all other site tests.
- (2) The chosen test procedure shall not jeopardize healthy parts of the substation.
- (3) The following parts may be disconnected for the test, either because of their high charging current or because of their effect on voltage limitation :
 - a. Overhead lines.
 - b. Power transformers and voltage transformers.
 - c. Surge arresters.

If the disconnection of the above parts are required, a removable link shall be included in the design of each of the above cases. The design shall be such as to minimize the risk of introduction of faults during the connections after the tests.

- (4) The high-voltage tests shall include the following tests:
 - a. AC voltage tests.
 - b. Lightning impulse voltage tests.
 - c. Switching impulse voltage tests.

The test voltage shall be 80% of the voltage specified for the respective routine tests. The AC voltage test shall be a duration of one (1) minute at the specified level. The lightning and switching impulse tests shall each consist of three (3) impulses of each polarity at the specified level.

6. QUALITY ASSURANCE/QUALITY CONTROL

The quality assurance/quality control of the substation, including the circuit breakers and other primary electrical equipment, shall conform to the requirements included in Part 4 Section 2 Subsection 2.7. I include, but not be limited to, the following:

- (1) All the tests, routine tests and any special tests performed on the equipment.
- (2) Special precautions taken at the factory to ensure a clean assembly.

- (3) Field installations procedure to ensure a clean assembly.
- (4) All proposed site tests to verify the quality of the equipment.

The documentation as specified shall include, but not be limited to, the following:

- (5) Reports of the type tests, routine tests and any other special tests performed at the factory.
- (6) Reports of all the site tests, including the test procedures, any failures and remedial actions taken.

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